



Brasília DF, 01 Agosto 2019

Going fast with G



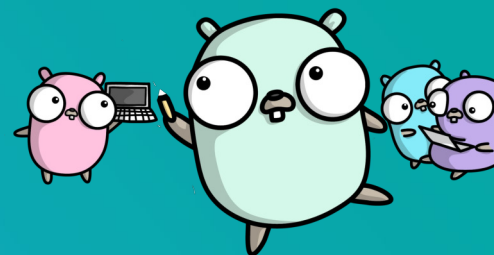
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GO Evangelist

SECTION ONE

Introduction



Go

For whom is the Go programming Language ?

- **Suitable for anyone, from beginners to experienced programmers.**

Why?

- **Easy to learn, and become productive**
- **Concepts are easy to understand**
- **Type and Memory Safe, Enforces Code Quality**
- **Modern Toolchain, fun to write!**

For what Go should be used ?

- Micro Services, Web, Networking, Infrastructure, CLIs, Data Science, Cloud Computing, Cross Platform Apps

Why ?

- Built for the Cloud
- Easy to maintain
- Stable Syntax since 1.0 (2012)
- Resource efficient and high performance
- Supported by big companies (e.g. Google)

HOW IT RUNS

- Compiled to machine code: 32 e 64-b
x86, ARM, RISC-V, WebAsm...
- Cross Platform: Linux, Windows, Mac,
Browser, IOS, Android, Arduino and ot
Microcontrollers (tinygo) ...
- Cross Compilation (for any OS/Arch)

CODE QUALITY AS A MAIN CONCERN

- **Statically Typed:** zero variable type related runtime errors
- **Garbage Collected: safe** for any programmer
- Born with fixed **Code Styling** and **Formatting Tools** (GO fmt)
- Powerful **Tests** and **Benchmarks** are built in the language

SECTION TWO

Language Design

Authors:

Rob Pike (Bell Labs Plan 9 OS, Unix)

Ken Thompson (C Language, Unix, UTF-8)

Robert Griesemer (Java HotSpot, V8 JS Eng)

GO VALUES

Thoughtful
Simple
Efficient
Reliable
Productive
Friendly

Deliberate and considerate.

Clear and precise.

Do more with less.

It just works.

Realize your vision, faster.

Accessible and welcoming.

DESIGN PRINCIPLES

Simplicity



All features
should be easy
to understand

DESIGN PRINCIPLES

Orthogonality



All features
should interact
in predictable
and consistent
ways.

DESIGN PRINCIPLES

Legibility

- Simple understanding without over verbosity
- Built with concepts familiar to any C or Java Developer

SECTION TREE

Dependencies Management

packages & module

Go packages

Always know where your dependencies come from

```
package main

import (
    "bytes"
    "context"
    "fmt"
    "log"
    "math/rand"
    "producer/db" // pacote interno

    "github.com/google/uuid"
    kafka "github.com/segmentio/kafka-go"
)
```

Packages are
imported using
the repository
URL

Go Modules

Arquivo de dependencias go.mod

```
module kafka_producer

go 1.12

require (
    github.com/google/uuid v1.1.1
    github.com/segmentio/kafka-go v0.3.0
)
```

```
module rabbitmq_producer

go 1.12

require (
    github.com/streadway/amqp v0.0.0-20190404075320...
)
```

Sem-Ver b
on reposit
releases

Want a spe
commit?
Commit Ha

SECTION FOUR

TOOLING

**A Modern and Powerful language should
have Modern and Powerful tools**

Go fmt

GO code looks familiar wherever you GO

```
package main
import ("log")
func main(){helloMessage:="Hello Go"
log.Println(helloMessage)}
```

```
$ go fmt test.go
```

```
package main

import (
    "log"
)

func main() {
    helloMessage := "Hello Go"
    log.Println(helloMessage)
}
```

Auto code
formatting
directly from
the toolchain

GO GET

GO RUN

GO BUILD

GO INSTALL

Want a library ?

```
go get github.com/google/uuid
```

```
go get github.com/google/uuid@latest
```

```
go get github.com/google/uuid@v1.0.0
```

GO GET

GO RUN

GO BUILD

GO INSTA

**Want to run with
one command ?**

go run hello.go

A solid teal bar at the bottom of the page, transitioning from a darker teal on the left to a lighter teal on the right.

GO GET

GO RUN

GO BUILD

GO INST

Want to build your app?

go build app.go

Want to Cross Compile ?

GOOS=windows GOARCH=amd64

go build ap

GOOS=linux GOARCH=arm

go build ap

gomobile build -target=android

app.go

GO GET

GO RUN

GO BUILD

GO INSTALL

Want a tool in your path ?

go install github.com/uber/kraken

Binary goes to:

- \$GOPATH/bin/kraken

In my case: /home/auyer/go/bin/kraken

Or

- /usr/local/bin/kraken If GOBIN=/usr/local/bin

goimports

```
package main

import (
    "fmt"           // fmt unused, will be removed
                   // log needed, will be added
                   /* github.com/streadway/amqp needed,
                      will be added if found in your system,
                      error if not recognized
                   */
)

func main() {
    url := "amqp://amqp.local"
    conn, err := amqp.Dial(url)
    if err != nil {
        log.Fatalf("cannot dial: %v: %q", err, url)
    }
    conn.Close()
}
```

Automatic
insertion and
removal of
imported
libraries

CODE QUALITY CHECK WITH Go Report Card

```
go get github.com/gojp/goreportcard
```

```
goreportcard-cli -v
```

```
Grade: A+ (99.9%)
Files: 332
Issues: 2
gofmt: 100%
go_vet: 100%
gocyclo: 99%
gocyclo download/download.go:22
    warning: cyclomatic complexity 17 of function
download() is high (> 15) (gocyclo)

golint: 100%
ineffassign: 100%
license: 100%
misspell: 100%
```

Run several
static check
tools to keep
your code in
good shape

Web Version :
<https://goreportcard.com>

Go test

Test files should be in the same package of the functions, and are called package_test.go

```
package pow

import (
    "math"
    "testing"
)

func TestMathPow(t *testing.T) {
    res := math.Pow(2, 2)
    if res != 4 {
        t.Fail()
    }
}

func BenchmarkHandler(b *testing.B) {
    for n := 0; n < b.N; n++ {
        math.Pow(2, float64(n))
    }
}
```

BenchmarkPow 20000000 113 ns/op 0 B/op 0 allocs/op

go test .

go test -rac

go test -be

SECTION FIVE

Features

SCALABLE MULTI-CORE PROCESSING

**Concurrency
with
Goroutines**

```
go doSomething()
```

- Lightweight,
- Scalable,
- Simpler Than
Thread

Communication between goroutines

'Read-exactly-once' communication channels

```
package main

import ("fmt")

func helloName(name string, resultsChannel chan string) {
    resultsChannel <- "Hello " + name + " !"
}

func main() {

    channel := make(chan string)
    go helloName("Gopher", channel)
    result := <-channel
    fmt.Println(result)
}
```

<https://play.golang.com/p/BmLWIGDNwfC>

*“Do not
communicate
sharing mem
instead, shar
memory by
communicati
– Rob P*

How many lines make a HTTP server?

```
package main

import (
    "fmt"
    "log"
    "net/http"
)

func função_olá(w http.ResponseWriter, req *http.Request) {
    fmt.Fprintf(w, "Olá !")
}

func main() {
    http.HandleFunc("/", função_olá)
    err := http.ListenAndServe("localhost:8080", nil)
    if err != nil {
        log.Fatal(err)
    }
}
```

Concurrent
Non-Block

Production
Ready with
stdlib only

<https://play.golang.org/p/G>

ABSTRACTION THROUGH INTERFACES

You can define methods on any type/struct:

```
type geometry interface {  
    area() float64  
    perim() float64  
}  
  
type rect struct {  
    width, height float64  
}  
  
func (r rect) area() float64 {  
    return r.width * r.height  
}  
  
func (r rect) perim() float64 {  
    return 2*r.width + 2*r.height  
}  
  
func measure(g geometry) string {  
    return fmt.Sprintf("Area: %f, Perim: %f", g.area(), g.perim())  
}
```

Interfaces
collections
method
signatures

Source:
<https://gobyexample.com>

SECTION SIX

Who uses it

Notable Companies Using GO

Adobe, Alibaba, AT&T, Atlassian, Booking.com, Cannonic, Cabify, CircleCI, Cloudflare, Comcast, Dell, Digital Ocean, Dropbox, Facebook, Google, IBM, Intel, Microsoft, Mozilla, PayPal, Pivotal, Twitter, Uber, SpaceX...

What about Brazil ?

99, MercadoLivre, Catho, C6 Banck, Dafiti, Elo 7, Globo.com, LEVEE, Loggi, Hotel Urbano, Viva Real, Magazine Luiza, M Rede Bandeirantes, PagSeguro, PMMG, Quinto Andar, SiBBR.gov.br, SumUp, TOTVS, Vórtx, Walmart Brasil ...

Fontes: <https://github.com/golang/go/wiki/GoUsers> e <https://www.meetup.com/golangbr>



Docker

Docker enables developers and IT operations to build, secure and manage applications without technology or infr

San Francisco, CA <https://www.docker.com> info@docker.com Verified

Repositories 137 Packages People 46 Projects 2

Find a repository... Type: **All** Language: **All**

docker-ce

Docker CE

git docker golang moby

Go 794 3,073 0 1 Updated 1 hour ago

engine

Forked from moby/moby
docker engine release repository fork of moby

Go Apache-2.0 15,679 299 0 27 Updated 2 hours ago

cli

The Docker CLI

docker cli

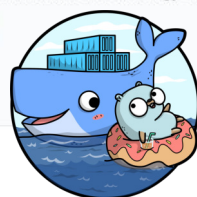
Go Apache-2.0 797 1,559 326 (3 issues need help) 122 Updated 5 hours ago

app

Make your Docker Compose applications reusable, and share them on Docker Hub

docker kubernetes helm applications compose

Go Apache-2.0 114 1,207 41 9 Updated 7 hours ago



Top languages

Go Shell PHP

Most used topics

docker cloud
docker-compose

People



Kubernetes

Kubernetes

<https://kubernetes.io> Verified

Repositories 69 Packages People 473 Projects 27

Pinned repositories

kubernetes

Production-Grade Container Scheduling and Management

Go 56k 19.5k

enhancements

Features tracking repo for Kubernetes releases

Go 729 425

website

Kubernetes website and documentation repo:

HTML 1.5k 5.2k

test-infra

Test infrastructure for the Kubernetes project.

Go 1.5k 963

Find a repository... Type: **All** Language: **All**

test-infra

Test infrastructure for the Kubernetes project.

k8s-sig-testing

Go Apache-2.0 963 1,455 287 (35 issues need help) 98 Updated 1 minute ago

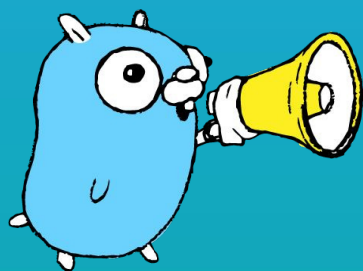
kubernetes

Production-Grade Container Scheduling and Management

containers kubernetes go cncf

Go Apache-2.0 19,487 55,972 2,159 (60 issues need help) 1,066 Updated 7 min ago

nado problem detector



Synthesis

Efficient, reliable, scalable, simple.

Easy to learn. Familiar Concepts.

Code Quality at its core

Strict rules = Easy to maintain

High performance with low footprint

It's growing, and it's not going away

TIME FOR A CHALLENGE



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Thank you for your time



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